Homework 6 **Brandon Aikman**

1. A two-way set associative cache has lines of 16 bytes and a total data size of 8 kbytes. The 64-Mbyte main memory is byte addressable. Show the format of main memory address in terms of tag, index and word. (10 pts)

Tag	Index	Word
25..12	11..4	3..0

2. For the hexadecimal main memory address 111111, 666666, BBBBBB, show the following information, in hexadecimal format, assume there are 16K blocks in the cache, each block has 4 words. (5 pts each; 15 pts total)
(Assuming padding left with 0s to reach multiple of four bits)

   (a) Tag, index, and word value for a direct-mapped cache.

111111:

Tag	Index	Word
0x11	0x0444	0x1

666666:

Tag	Index	Word
0x66	0x1999	0x2

BBBBBB:

Tag	Index	Word
0xBB	0x2EEE	0x3

   (b) Tag and word value for a fully associative cache.

111111:

Tag	Word
0x044444	0x1

666666:

Tag	Word
0x199999	0x2

BBBBBB:

Tag	Word
0x2EEEEE	0x3

   (c) Tag, set, and word value for a two-way set associative.

111111:

Tag	Set	Word
0x022	0x0444	0x1

666666:

Tag	Set	Word
0x0CC	0x1999	0x2

BBBBBB:

Tag	Set	Word
0x177	0x0EEE	0x3

3. Consider a machine with a byte addressable main memory of 64k bytes and block size of 8 bytes. Assume that a direct mapped cache consisting of 32 lines is used with this machine. (5 pts each; 15 pts total)

   (a) Into what line would bytes with each of the following addresses be stored?
   0001 0001 0001 1011 - **3**
   1100 0011 0011 0100 - **6**

1101 0000 0001 1101 - **3**
1010 1010 1010 1010 - **21**

(b) Suppose the byte with address 0001 1010 0001 1010 is stored in the cache. What are the addresses of the other bytes stored along with it?
0001 1010 0001 1000
0001 1010 0001 1001
0001 1010 0001 1010
0001 1010 0001 1011
0001 1010 0001 1100
0001 1010 0001 1101
0001 1010 0001 1110
0001 1010 0001 1111

(c) How many total bits of this cache. (including a validation bit for each cache line)
2336 bits. (64 data + 8 tag + 1 validation) * 32 lines.

4. Consider a cache of 4 lines of 16 bytes each. Main memory is divided into blocks of 16 bytes each. That is, block 0 has bytes with address 0 through 15, and so on. The Least Recently Used replacement scheme is used. Now consider a program that access memory in the following sequence of addresses:
Once: 63 through 70
Loop ten times: 15 through 32; 80 through 95 (20 pts)

(a) Suppose the cache is organized as direct mapped. Compute the hit ratio.
324/348 = 93.1%

(b) Suppose the cache is organized as two-way set associative. Compute the hit ratio.
342/348 = 98.3%